Delivery Time Prediction for Olist E-commerce Platform

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Executive Summary

The project aims to develop a prediction model to estimate the average delivery time of the Olist e-commerce platform to support data-driven logistics optimization and customer satisfaction improvement. Through systematic exploratory data analysis, we have identified the key factors affecting the delivery time, including geographical distance, cross-state status, product characteristics and transportation costs.

During the data preprocessing stage, we cleaned and integrated over 100,000 order records and constructed key variable features to better capture nonlinear relationships. We removed extreme outliers (e.g., delivery time, geolocation, and product volume) and applied winsorization to cap variables such as total price and freight. Missing values were filled with median values to enhance model robustness. Moreover, several new features were engineered based on logistics logic, including 'is_cross_state' and 'customer_seller_distance_km', capturing the impact of inter-state shipping and distance on delivery time. We also applied target encoding to categorical features (e.g., customer_state_mean_delivery, category_mean_delivery), enabling the model to benchmark regional and product-level efficiency. In addition, log-transformations were used to reduce skewness in variables such as freight, volume, weight, and payment value, allowing the model to focus on the normal delivery cases.

After evaluating a variety of algorithms (Linear, Lasso, Random Forest, LightGBM and XGBoost), both the optuna-tunised XGBoost and LightGBM obtained the best benchmark performance. Finally, a stacked integration model (Base: XGBoost & LightGBM, Meta:LightGBM) was implemented, and MAE = 3.6457, RMSE = 6.3413 was realized on the test set. The model takes into account predictive performance and interpretability and can stably identify key influencing factors.

The study found that logistics distance and regional differences are the main driving factors affecting the delivery time of Olist. Based on this prediction system, Olist can provide users with a more accurate estimated arrival time at the ordering stage and optimize the allocation of transportation resources, so as to improve customer satisfaction and reduce the risk of delay.

Overall, the project has successfully established a practical predictive-driven logistics analysis framework, which provides a scalable technical foundation and commercial value for Olist's data-driven operations.

1. Problem formulating and Business understanding

1.1 Business context

As one of the largest e-commerce platforms in Brazil, Olist connects many small sellers with consumers across the country, and its overall operational efficiency is largely dependent on the performance of

logistics services. Late delivery or large fluctuations in delivery time will directly cause business risks at multiple levels, including:

- Reduced customer satisfaction - Research shows that in e-commerce and express delivery services, when packages are not delivered on time, or when the expected delivery time is not realistic, it will seriously undermine customer trust in the platform.
- Increased return rate and complaint rate - Delivery delays directly affect users' shopping experience, which may lead to more returns and increased after-sales support costs.
- Reduced resource scheduling efficiency (involving warehousing, manpower, customer service, etc.) - the uncertainty of logistics links will lead to resource allocation failure, often requiring additional costs to deal with emergencies.
- Platform reputation damage and repurchase rate decline - poor delivery experience will damage users' overall impression of the platform, which will affect customer loyalty and repeat purchase intention in the long run.

Providing more accurate estimated delivery date (EDD) can bring significant positive impact to e-commerce platforms (Masuch et al., 2024). If Olist can make a reasonable prediction of the expected delivery time in the "pre-order" stage of customers, the platform will have the opportunity to identify potential "slow orders" in advance, to actively adjust the storage strategy, reasonably allocate logistics resources and optimize the distribution path. This will not only help the platform maintain its advantage in the increasingly fierce e-commerce competition but also bring consumers a more reliable shopping experience. Therefore, for Olist, predicting the "average delivery time of each order" in advance not only has clear commercial value, but also reflects the platform's emphasis on user experience. This predictive ability will not only drive operational efficiency, but also enhance customer satisfaction, reduce unnecessary costs, and ultimately increase user repurchase intention.

1.2 Business problem definition

In this context, we specify the following business problem: How can Olist predict the average delivery time of an order from order placement to customer receipt based on variables such as customer attributes, product information, seller characteristics, payment method, and order time for each order? This problem covers a few key dimensions:

- The unit of analysis takes orders as the basic analysis objects.
- Target variable: delivery time days, which is the difference in days between the actual delivery date of the customer and the order placing date.
- Input characteristics: cover customer location information (such as zip code, city, state), order time characteristics, product structure attributes (such as number of goods, weight, volume, category), seller's location and delivery ability, and payment related information (payment type, amount, etc.).
- Business application scenario: The prediction will be made in the post-order generation. The platform can adjust operational resources accordingly, such as selecting the optimal warehouse, prioritizing potential slow orders, or dynamically pushing estimated delivery times to customers, thereby improving overall service transparency and responsiveness.

1.3 Justification of Machine Learning

Based on business requirements and data conditions, it is reasonable to use machine learning method to predict EDD. First, the delivery time is complicated by multi-dimensional factors such as geographical distance, seller efficiency, commodity attributes, order placing time, payment method and logistics capacity, and there are many nonlinear relationships and interaction effects among these variables. Traditional statistical methods are difficult to fully capture such complex patterns, while machine learning models have significant advantages in processing high-dimensional features and identifying potential rules. Existing research in the field of logistics has also confirmed that its predictive performance is superior to traditional methods (Rokoss et al., 2024). Second, the rich historical data accumulated by Olist platform provides a solid foundation for machine learning. This technique supports models that are continuously optimized with new data feedback and can adapt to dynamic changes in the business environment.

1.4 Business Success Metrics

The value of this model will be reflected in the actual improvement of the following key business indicators: logistics efficiency, customer satisfaction, operating costs and return rate. The predictive insights output by the model are the starting point of optimization, and the subsequent key steps lie in quantifying the specific value created by these insights after guiding operational interventions through rigorous evaluation.

2. Exploratory Data Analysis (EDA)

2.1 Data Preparation and Integration

The Olist dataset is composed of multiple interrelated tables from the Olist e-commerce platform. Through a structured data integration and transformation process, the raw tables were merged and harmonized into a single order-level analytical dataset suitable for predictive modeling.

The process involved:

- Cleaning and standardizing timestamps to calculate the target variable.
- Dropping missing values and removing extreme outliers to improve data reliability and model robustness.
- Aggregating item-level details to the order level.
- Deriving categorical features using mode-based aggregation.
- Integrating payment information to capture total value and installment behavior.
- Computing geographical distances between customers and sellers using ZIP-code-based geolocation data and the Haversine formula.

2.2 Dataset Overview

Aspect	Description
Total records	96,462
Total columns	29
Data types	11 categorical / 6 numerical / 4 datetime
Target variable	delivery_time_days – Number of days between purchase and delivery
Key features	price_sum, freight_sum, product_weight_g_sum, product_volume_cm3_sum, customer_seller_distance_km, payment_installments_max

Following data cleaning and integration, key derived variables were generated to support subsequent analysis. The resulting dataset is summarized below.

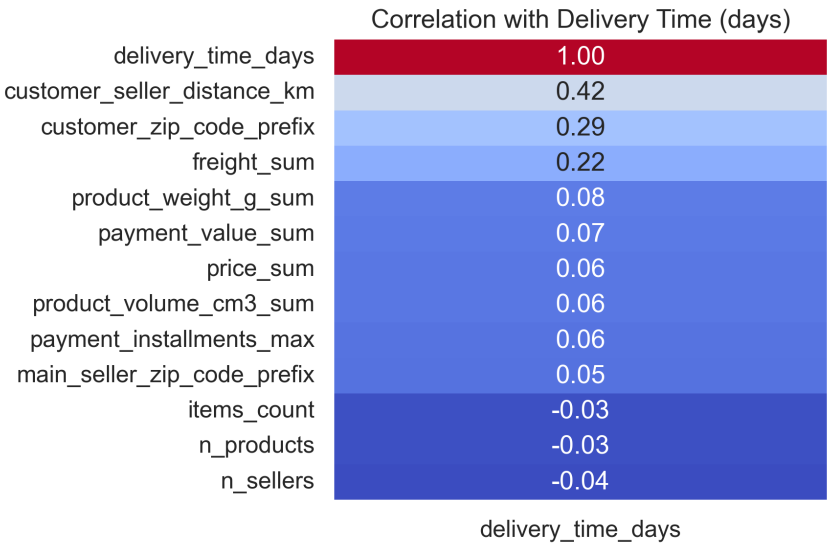
2.3 Descriptive Statistics and Visual Insight

2.3.1 Correlation Analysis

To identify the key drivers of delivery performance, a correlation analysis was conducted between the target variable and a set of numerical predictors at the order level.

Figure 1 summarizes the pairwise correlations, revealing that the geographical distance between customers and sellers exhibits the strongest positive correlation with delivery time ($r = 0.42$). This relationship is intuitive, as greater spatial separation naturally increases transportation complexity and transit duration.

Among the logistics-related variables, freight cost demonstrates a noticeable positive correlation, implying that higher shipping charges are often associated with longer or more complex deliveries. Similarly, product weight and product volume show smaller but meaningful positive relationships, indicating that larger or heavier packages tend to require extended handling and delivery times.
Conversely, transactional and financial variables such as order price, payment value, and installment count display only weak correlations ($r < 0.1$) with delivery duration. This finding suggests that the economic value of a transaction is not a direct determinant of delivery time, reinforcing the importance of focusing on logistical and geographical attributes in modeling.

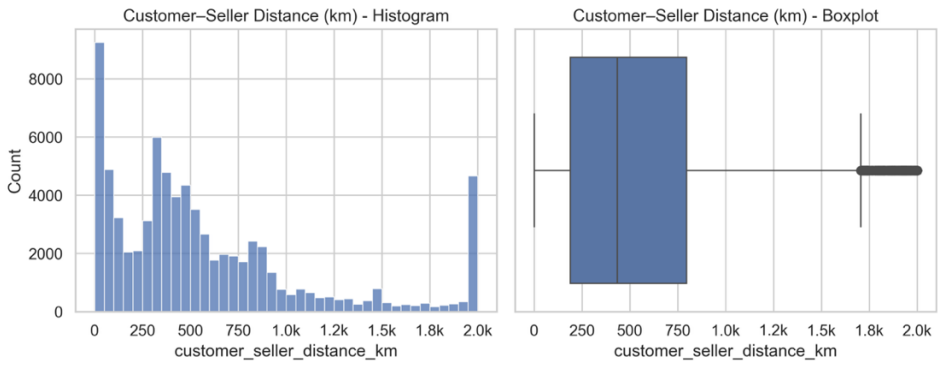


(Figure 1)

2.3.2 Distribution and Outlier Analysis for key variables

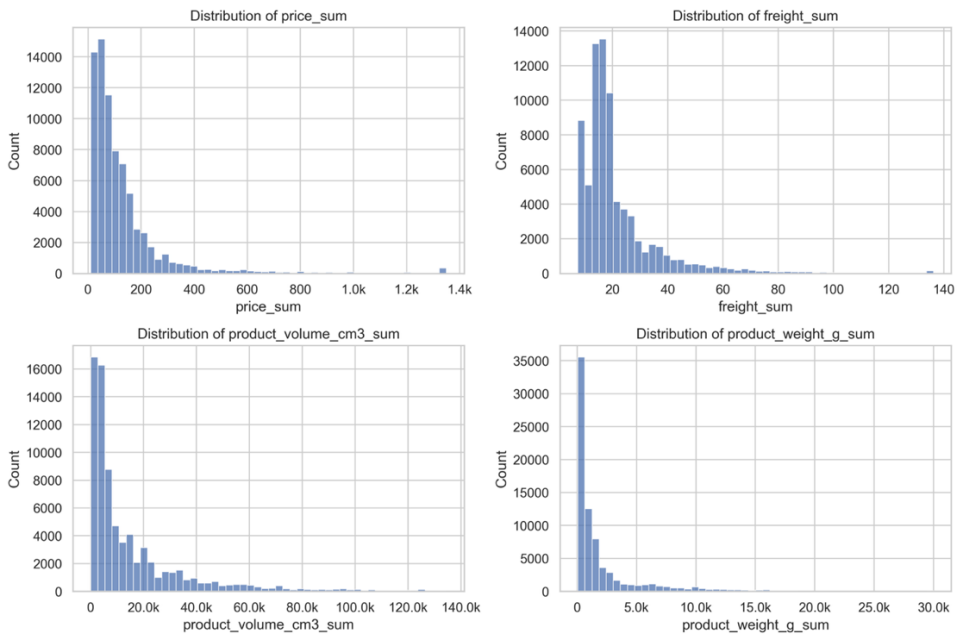
Following the correlation analysis, the key predictors most strongly associated with delivery time were examined to understand their underlying distributions and detect potential outliers.

Figure 2 presents the distribution of customer–seller distance. Most orders are concentrated within short- to medium-range deliveries (below 500 km), with a sharp decline in frequency as distance increases. The data show a right-skewed distribution, characterized by a minor cluster of long-distance deliveries (approximately 2,000 km). The corresponding boxplot （Figure3） confirms the presence of numerous high-distance outliers, suggesting substantial variability in delivery range.



(Figure 2-3)

Figures 4-7 display histograms for four shipment- and cost-related variables. All variables exhibit highly right-skewed distributions, a common pattern in e-commerce transactions. Most orders involve relatively small packages with low prices, light weights, and modest freight costs, while a limited number of high-value or heavy shipments form long right tails. These extreme observations likely represent large household appliances or long-distance deliveries with elevated handling and transportation costs.



(Figure 4-7)

2.3.3 Data Quality and Bias Review

A detailed examination of numerical features reveals non-trivial proportions of extreme values, indicating significant outliers across several variables. Specifically:

Variable	**Outlier Ratio**	**Interpretation**
main_seller_zip_code_prefix	**15.6%**	Geographic identifiers with unusually high or low frequencies, potentially reflecting regional concentration or data-entry inconsistencies.
product_weight_g_sum	**13.7%**	Heavy-tailed distribution, likely driven by a subset of multi-item orders that disproportionately affect overall weight statistics.
freight_sum	**9.7%**	High heterogeneity in shipping costs across transactions.
price_sum	**7.4%**	Substantial variation in order value, consistent with a wide range of basket sizes.
customer_seller_distance_km	**7.7%**	Right-skewed tail with a small cluster of long-distance deliveries (~2,000 km).
delivery_time_days	**5.0%**	A moderate share of unusually long or short delivery durations, possibly linked to logistic bottlenecks or regional infrastructure differences.

The observed skewness indicates that a minority of transactions contribute disproportionately to overall logistical effort. Consequently, this distributional asymmetry should inform model design, prompting the use of techniques that mitigate the impact of outliers and capture non-linear relationships between distance and delivery time. Building on these findings, our feature engineering focused on spatial drivers by deriving distance-related variables such as is_cross_state. Logarithmic transformations were applied to heavy-tailed predictors to reduce skewness. Among several tested algorithms, a stacked

ensemble combining XGBoost and linear regression achieved the best performance, effectively capturing both non-linear distance effects and interpretable linear trends identified in the EDA.

2.3.4 Linking EDA Insights to Olist's Business Objectives

Olist aims to reduce delivery uncertainty and improve operational efficiency by predicting order-level delivery times. EDA findings directly inform these goals:

Insight	**Business Relevance**
Distance and freight dominate delivery variance	Enables early detection of "slow orders" for smarter routing and warehouse allocation.
Long-tail order distributions	Supports segmentation between small/fast and large/slow orders for better resource planning.
Weak link with product or price	Confirms logistics, not product type, as the main delivery driver, suggesting focus on route and fulfillment optimization.
Outliers in freight and weight	Identifies high-cost deliveries where targeted interventions can maximize savings.

3. Feature Engineering

3.1 Preliminary Feature Engineering

After selecting main features during Exploratory data analysis, we aim to better prediction on average delivery time; hence, we move further analysis before building models. In data cleaning process, we create volume by multiplying height, length and width. We move some outliers such as geolocation, delivery time and volume of products during data cleaning (first step). Also, we use Winsorize to clip outliers like sum of price, sum of freight, etc. Also, we fill all of missing values to median (selected features only) as it's more robust for our prediction. We created some features based on operational and logistics logic to assist model to capture key factors of the changes of delivery time.

We created geographical and logistics features including 'is_cross_state', and 'customer_seller_distance' to reflect how whether cross state or not and distance influence delivery time in actual business. We use binary variable 'is_cross_state' to identify patterns of delivery time as it is significant in actual business. Although 'customer_seller_distance' is also important, whether cross state or not also includes logistics network connection and different distribution centers. For example, some logistics companies are only able to deliver parcel in the same state, and cross-state parcels may involve several logistics companies and distribution centers.

Moreover, we use Target encoding to transform categorical features, including 'customer_state_mean_delivery', 'seller_city_mean_delivery', 'category_mean_delivery', etc. They are operational benchmark, using average time of region and category to identify slow region and inefficient category and products, and supporting warehouse and route optimization.

We also use log-transformation based on highly right-skewness distributions. It would be better for us to capture volume, weight, payment value, and installments. They are product and payment level, aiming to reduce the impact on outliers and focusing more on normal orders to make predictions.

3.2 Further Feature Engineering

To improve our model for prediction, we changed our scale on test data and our prediction from delivery time to 'log delivery time'. In addition, we add some interaction features including interactions with freight and weight, items with volume, distance with cross-state or not, payment value and installment, volume and weight and delivery speed gap between customers and seller. According to their distribution, we only keep interactions with freight and weight, freight and weight and delivery speed gap. In Olist business, sometimes interaction effects affect average delivery time, and we explore these factors on how to impact our forecasting. To reduce the effects of outliers, we took log and delete other three interactions.

Based on the visualization, our target (delivery time) was also highly right skewed. Taking log would stabilize our model and focus on relative changes in delivery times, and we will use $expm1$ to convert log-transformation for meaningful business context. It will improve our business goal by identifying slow orders and optimizing logistics efficiently.

4. Recommended models & justification

4.1 Baseline model

We recommend linear regression (after preliminary Feature Engineering) as our baseline model with 5-Fold Cross-validation. For model construction, we could use a single model to analyze some potential trends and make improvements for further anlaysis in complex models. Also, it is easier to interpret the reason of prediction and verify main trend for our prediction. Based on the metrics in Appendix A1.2, it may not sensitive for some slow orders and extreme situations. The model may not perform well for some features; however, it can provide insights for our improvement and overall trend. For example, we can consider some non-linear model to capture more significant features and better performance for some slow orders. In addition, taking the cost for consideration, linear regression model uses low computation cost and is easy to maintain. It's suitable for prediction in the early stage for prediction.

4.2 Best single model

We select XGBoost with the best hyperparameters as our best single model. Since we captured the main information from the linear regression model, we tried one linear model (Lasso) and several non-linear models for analysis. Although based on Appendix A4.2, XGBoost is performing as the best model, and it's more stable for us to further analyze the trend of features.

When building models, we also build XGBoost model without hyperparameter turning. It is able to compare metrics intuitively, and XGBoost model with best hypermeter shows better performance. In addition, turning hyperparameter by optuna is able to find balance between bias and variance trade-off.

The model is better for analyzing features. According to Appendix A4.2, total freight and 'customer_city_mean' perform better, and they relate to cost of logistics and regional variations. They provide insights for Olist for operation. For computation cost, it's slower than LightGBM and linear model. However, it's much better than random forest since XGBoost performs faster under hyperparameter selection and Cross-validation.

Overall, XGBoost is stable and more interpretable based on comparison. It's crucial for Olist to consider forecasting.

4.3 Stacking Model

The "model stacking" scheme we recommend is a two-layer structure. The basic layer is operated in parallel by XGBoost and LightGBM, and the top layer is a LightGBM metamodel, which is responsible for integrating basic predictions and correcting deviations.

We recommend that its business logic is very clear: the average predicted deviation (MAE) of a single XGBoost model is 4.86 days. The stacking model significantly compresses this deviation to 3.65 days, and raises the R^2 value from 0.316 to 0.501.

We accept the price of it as a black box, because it is exchanged for an average extra 1.2 days of prediction accuracy. This income will directly improve customer satisfaction, and its commercial value far exceeds the cost of technology.

The technical performance and comparative evaluation of these models are further detailed in the following Model Evaluation section.

5. Model evaluation

5.1Performance Evaluation of the Three Models

This section systematically evaluates the performance of the three models, focusing on two key business issues: the average number of days of prediction error, and the model's understanding of delivery time fluctuations. Table below outlines the main evaluation indicators.

Model	MAE	RMSE	R2
Stacking (XGBoost + LightGBM + LightGBM Meta)	3.6457	6.3413	0.5013
XGBoost (Optuna+CV)	4.8611	7.4258	0.3161
Linear Regression (5-fold CV)	5.1386	7.7245	0.2600

- Baseline Model (Linear Regression): The average MAE is 5.14 days. From a business perspective, it is extremely risky to rely on such forecast results with an error of more than 5 days to formulate logistics plans or customer delivery commitments. Its decision coefficient R^2 is only 0.260, showing that the model is almost unable to explain the main factors behind the delivery delay.

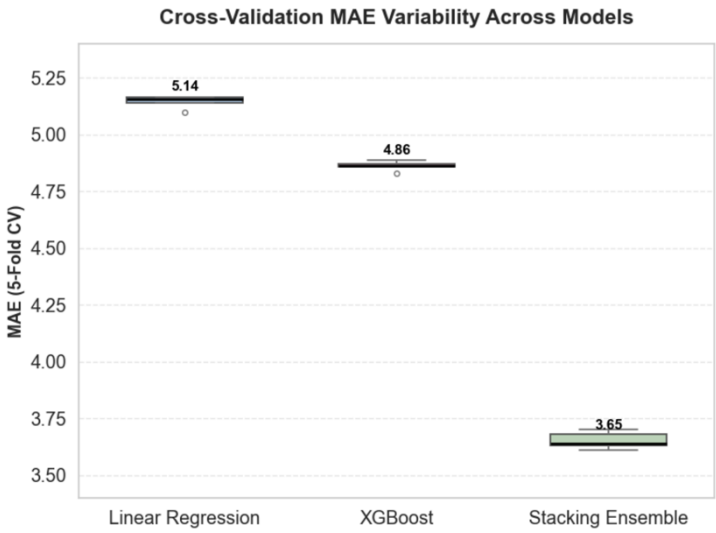
- Single model (XGBoost): The XGBoost model reduces the average error to 4.86 days. Although the accuracy has been slightly improved, the improvement at the business level is still quite limited, and the reliability of the forecast has not been significantly improved.
- Stacking model (XGBoost + LightGBM + LightGBM Meta): The stacking model showed significant advantages. The MAE was reduced to 3.65 days, about 1.2 days higher than XGBoost, and the prediction accuracy reached the applicable level for the first time. Its R^2 increased to 0.501, far exceeding 0.316, indicating that the model can explain more than half of the delivery time fluctuations, providing substantial support for understanding and optimizing logistics uncertainty.

The stacking model not only performs best in terms of technical indicators (the lowest MAE), but more importantly, it is the only model that can support business decision-making at present, providing a reliable basis for formulating more accurate delivery strategies.

5.2 Robustness & Generalization Analysis

This part aims to verify whether the outstanding results (MAE = 3.65 days) in Section 5.1 are truly reliable, not accidental.

We use 5-Fold cross validation to test the performance of the model on the training data set. The results shown in the figure show that the box distribution of all models is extremely compact, which shows that the model has not been fitting, and the overall performance is stable and repeatable.

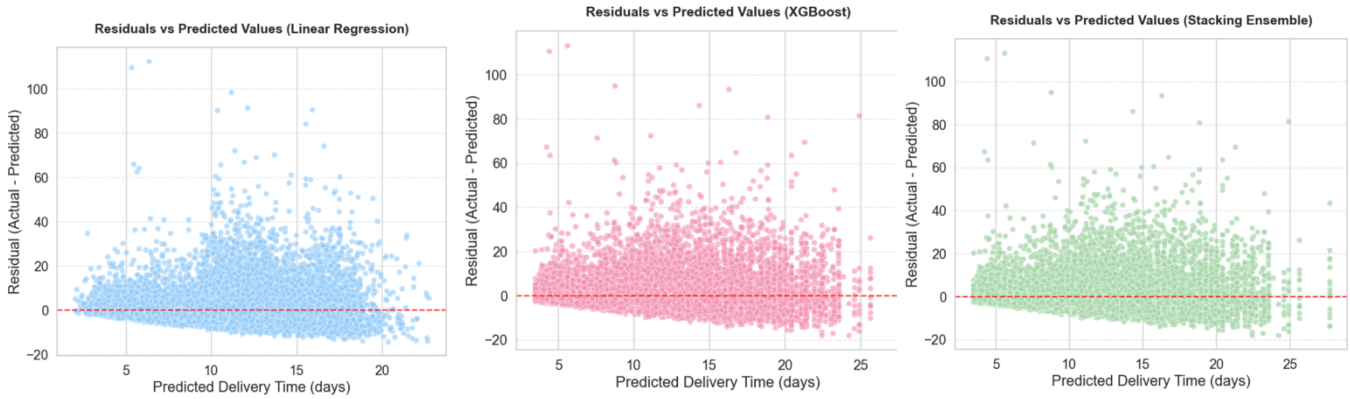


However, stability itself does not mean that the model has commercial value. Linear regression is manifested as stable and bad, and the error has always been maintained at about 5.14 days in all experiments. The Stacking model is stable and excellent, maintaining a low error of about 3.65 days in each verification, and the performance is highly consistent.

Overall, this box plot provides full confidence for business decision-making: the 3.65-day MAE of the Stacking model is not accidental, but a reliable and generalizable capability. This shows that the model is expected to maintain a high level of forecast performance continuously and steadily in the real business environment in the future.

5.3 Error Analysis

In order to evaluate the prediction bias and calibration of the model, this study drew a distribution chart of predicted value - residual. Residual (Y axis) is defined as actual delivery time − predicted delivery time. In the ideal state, the residual difference should be uniformly and symmetrically distributed on both sides of the 0 line (red line), and there should be no obvious structural trend.



- Linear regression model: The residual distribution is obviously upward, and a large number of sample points are concentrated above the 0 line, indicating that the model systematically underestimates the actual delivery time. As the forecast value rises, the residual discreteness expands, showing typical heterovariance characteristics, reflecting the insufficient fit of the model in the high prediction interval, and the forecast is generally too optimistic. From a business perspective, the model tends to provide short delivery expectations, increasing the risk of customer satisfaction.
- XGBoost model: XGBoost has a significant improvement in residual balance. The distribution of point clouds on both sides of the 0 line is more uniform, and the number of high residual samples is reduced, showing that the model can capture some nonlinear features. However, its dense area is still slightly above the 0 line, and there is still a slight undervaluation tendency. Overall, XGBoost has significantly reduced the systemic bias, but the forecast is still slightly optimistic.
- **BEST - Stacking model:** The residual distribution of the stacking model is the most compact and symmetrical, highly concentrated around the 0 line, with the least extreme error points, and the distribution is balanced in different prediction intervals. This shows that the model performs best in eliminating systematic deviations and can maintain prediction consistency and stability in a full range, reflecting excellent prediction calibration ability.

5.4 Business Implications

The results of the comprehensive performance evaluation show that the stacking model is superior to other models in terms of accuracy, calibration and stability. Its MAE is the lowest, R^2 is the highest, and it shows the most ideal symmetry distribution in the residual analysis, which almost eliminates the systematic undervaluation bias and makes the prediction results close to unbiased estimation.

At the business level, this means that the forecast value output by the model is highly reliable, and can be directly used for operational planning and customer commitment management. Enterprises can set a reasonable and transparent delivery time window (such as predicted value ± 3 days) according to the

model prediction results, which can significantly improve delivery controllability while maintaining the stability of customer experience.

At the same time, with excellent stability and generalized performance, the prediction results of the stacking model can be used as a unified cross-departmental decision-making benchmark, providing consistent and reliable data support for core links such as logistics scheduling, resource scheduling and performance evaluation, so as to strengthen the predictability and trust foundation of the overall operation.

6. Business Insights

This section combines the key analysis results of exploratory data analysis (EDA) and model evaluation, thereby extracting meaningful and actionable business insights. The objective is to transform research outcomes into strategically significant practices, achieve data-driven decision-making, and thereby enhance delivery reliability, operational efficiency, and customer satisfaction.

6.1 Key insights

Insights 1: Region-level and distance-related factors dominate delivery performance

According to the results of the correlation analysis, the distance between the seller and the customer has the highest correlation coefficient (0.42) with the delivery time. In addition, variables with regional average duration characteristics also have relatively high weights, indicating that there are differences in delivery speeds among different regions. This phenomenon reflects the potential problems of Brazil's vast territory and the uneven cross-state transportation infrastructure.

It can be seen from this that the delivery time is mainly affected by geographical location and infrastructure conditions. In contrast, the attributes of the goods or payment methods have little impact. Olist should prioritize focusing on geographic-based logistics optimization, such as optimizing cross-state routes, adjusting regional warehouse distribution, and dynamic carrier scheduling, thereby alleviating transportation issues between different regions.

Insights 2: The order distribution is highly skewed, and long-tail orders reveal potential operational inefficiencies.

Through the distribution analysis of key predictor variables such as distance, freight charges and commodity weight, we found that they generally exhibit significant right-skewed distribution characteristics. This phenomenon indicates that many orders on the platform fall within the regular range, but there are a few extreme cases, such as orders with a transportation distance exceeding 2,000 kilometers or a package weight over 20 kilograms. Although the proportion of such orders is small, they account for a relatively high proportion in the overall delivery delay and cost fluctuations. Therefore, the interaction variables introduced in the model (such as the logarithmic freight - weight interaction term and volume - weight interaction term) also verify the complex nonlinear relationship existing among freight volume, weight and delivery time.

From a business perspective, these "slow orders" or "high-risk orders" distributed at the long tail end constitute a key risk group that cannot be ignored in Olist's operation and management. Therefore, identifying such orders in advance and continuously monitoring them can help the platform more accurately schedule logistics resources and plan routes more efficiently, thereby effectively controlling the risk of timeout, reducing unexpected costs, and improving overall operational efficiency and customer satisfaction.

Insights 3: Transaction and financial variables have a limited impact on estimated delivery time

We also found that financial variables such as order amount, total payment amount and number of installments have very limited impacts on delivery timeliness (correlation coefficients are less than 0.1, and the model gain weights are generally low as well). This discovery indicates that the key factor determining distribution efficiency does not lie in the economic value of orders, but rather more in the complexity of the logistics supply chain and geographical dimensions.

This discovery suggests that Olist should shift its strategic focus from the traditional "price management" mindset to the more fundamental "supply chain collaboration" capability building. To effectively shorten the delivery cycle and reduce operational uncertainty, the platform should focus on investing in the intelligent construction of the logistics system, the establishment of a predictive monitoring system, and the optimization of the seller's shipping efficiency, rather than overly relying on financial incentives or price control measures.

6.2 Actionable recommendations

Based on key insights from data analysis and model output, we propose the following three interrelated strategic suggestions for the Olist platform to systematically enhance the efficiency of logistics operations.

First, build a distribution risk scoring system to achieve predictive logistics management. It is suggested that the key features identified by the model be utilized to develop a set of "slow order risk index", which can identify and mark potential delayed orders at the order generation stage. It can trigger intervention through proactive early warning and effectively reduce the risk of customer complaints.
Second, implement a stratified logistics strategy based on delivery distance and state boundaries. It is suggested that orders be divided into two levels for differentiated processing: adopt the standard delivery plan for short-distance (intra-state) orders; For long-distance or cross-state orders, the timeliness is guaranteed by setting up dedicated routes and establishing regional distribution nodes. This hierarchical model can significantly enhance cost control and resource utilization efficiency while maintaining service quality.

Third, strengthen the operation analysis system at the regional and seller dimensions. The platform should establish regional and seller performance benchmarks based on this, accurately identify inefficient units, and continuously promote the overall efficiency improvement of the logistics network through targeted training, performance incentives and dynamic feedback mechanisms.

6.3 Limitations and assumptions

Although this study has reliable evidence in terms of statistical significance and predictive performance, there are still several limitations that need to be explained. Firstly, the data used in the research mainly comes from historical transaction records from 2017 to 2018, which fails to fully reflect the potential changes in logistics infrastructure and consumption patterns after the pandemic. Secondly, what the analysis reveals are the predictive relationships among variables rather than causal mechanisms - for instance, although distance and cross-state distribution show significant performance in the predictions, their effects may not directly equate to the actual effects of policy intervention. Furthermore, other factors that are theoretically highly correlated with the speed of delivery, such as traffic conditions, weather conditions and holiday fluctuations, were not included in the model, which may lead to some variances not being fully explained. Meanwhile, there are certain deviations in the prediction effect of the model. During the model experiment, we found that due to the differences in the programming environment used by different members, the model will have slight numerical deviations, such as the slight fluctuations of MAE or RMSE. However, these gaps are relatively small overall and will not affect the overall trend of the model, the ranking of feature importance and the stability of the main conclusions.

Looking ahead, subsequent research can introduce causal inference methods, such as difference-in-differences models, to more accurately identify the true effects of various logistics optimization measures, and further integrate real-time tracking data, thereby enhancing the model's adaptability and explanatory power in dynamic operational environments.

7. Business implementation and impact

7.1 Deployment Strategy

The proposed stacking model, based on XGBoost and Linear Regression, will be implemented as a key predictive service in Olist's order-management ecosystem in order to predict the delivery time for new orders at the moment of purchase. This will allow the company to foreseen possible delays, improve resource allocation and enhance customer satisfaction. From the technical perspective, the system will be implemented as a microservice connected by API to Olist's data warehouse. In the case of new orders, input features, such as distance from customer to seller, freight value, product weight and volume of shipment, will be automatically extracted from transactional databases and sent to the predictive API. The latter will return the predicted delivery length in days, which will be written back in the operational data and shown in dashboards that can be consulted in Olist's main business units. Moreover, the system will be implemented using a modular architecture to ensure scalability and maintainability. A minimum daily ETL process will refecth the input data, while quarterly retraining periods will include the most recent orders in order to maintain adequate model behaviour and react to logistics variations.

In practical terms, the system will be input by three different departments:

- Logistics and Warehousing: for the prioritisation of shipments and for the optimisation of routing based on predicted delivery dates.

- Customer Service: to ensure that customer have accurate and transparent estimated delivery dates, while maintaining good expectations when delays are foreseen.
- Data Analytics and Operations: in order to monitor model performance, as well as maintain the infrastructure of the API and coordinate retraining activities.

With this idealised deployment system, the immediate and quantifiable predictive outputs will be transformed into actionable insights in terms of activities across the Olist value-chain, where data-driven predictive will be closely linked to operational decision-making and delivery outcomes.

7.2 Risks, Challenges, and Mitigation Strategies

The adoption of a predictive analytics system on a large scale will inevitably bring about technical, operational and ethical challenges. For Olist, the effectiveness of the suggested stacking model depends on its not only predictive accuracy, but also its long-term reliability, integration stability and fairness in business practices. Thus, it is essential that key risks be identified proactively and clear mitigation mechanisms put in place to guarantee sustainably the performance of the model and the alignment with business.

Risk / Challenge	Description & Potential Impact	Mitigation Strategy
Model Drift	The underlying feature distributions (e.g., distance, freight cost) may evolve due to new logistics routes, changing customer patterns, or seasonal demand, leading to decreased predictive accuracy.	Monitor model performance metrics (MAE, RMSE) via automated dashboards. Trigger retraining when performance deviates > 10% from baseline or at least once per quarter.
Data Quality & Integration	Incomplete, delayed, or inconsistent order records could distort predictions or disrupt API responses during integration with the order-management system.	Implement automated data validation and imputation procedures; conduct pilot deployment in selected regions before full-scale rollout.
Operational & Maintenance Challenges	Limited technical resources or unclear ownership may cause insufficient model monitoring or delayed updates, reducing reliability over time.	Assign the analytics team responsibility for ongoing maintenance and documentation. Establish quarterly model review meetings with logistics and operations units.
Ethical & Fairness Considerations	The model may inadvertently generate biased predictions that disadvantage specific regions or sellers, undermining trust and platform equity.	Perform periodic bias audits segmented by geography and seller type. Maintain transparent reporting of model outcomes and corrective adjustments.

Taken together, these steps create a governance framework for the model that guarantees to provide for reliability, accountability and fairness throughout the model lifecycle. With this combination of automated checking and human oversight, Olist can also be guaranteed that its predictive system can provide true, fair and value-adding information throughout the changing market conditions.

7.3 Monitoring and Continuous Improvement

To ensure continuous model reliability and business value over time, a continuous monitoring and improvement framework will be implemented. This framework combines automated performance tracking, business-relevant evaluation metrics, and periodic retraining to ensure the model remains up to date with the changes in Olist's operating environment over time.

Technical Monitoring

The analytics team will monitor a real-time dashboard that looks at key model statistics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. Distributions of input features will be looked at as well to detect data drift or anomalous patterns in input features. Alerts will be generated anytime model performance begins exhibiting a significant difference in performance (defined here as greater than 10% increase in MAE in relation to the base performance metrics). They will retrain the model at that point and use the last three months of data for retraining. These trainings will be executed at least quarterly, even if performance checks do not yield warning signs, since operational conditions and customer behavior will change over time.

Business KPI Monitoring

Furthermore, in addition to the technical indicators, the effectiveness of the models will be measured against key business Key Performance Indicators (KPIs): Percentage of on time delivery - percentage of supply orders completed in the predicted time; Average delay in delivery. Difference in predicted and actual delivery time; Customer complaints percentage. Amount of complaints with respect to logistics detractors. These measures will be expressed in a common dashboard that is available to the logistics and customer service teams, reinforcing the transparency of data science with respect to business matters.

Continuous Improvement Cycle

To examine the insights from the models, find new predictors, and ascertain if retraining or changes in predictors are necessary, quarterly review meetings and discussions of the modelling process will occur. Controlled A/B tests of new model versions will be run prior to full implementation to ascertain the improvements in performance of the models. The continuing feedback of this process allows for the predictive analytics to be adaptive, interpretable, and strategic in nature, changing the model from a static tool to a learning system which tracks the changes in Olist's logistics process.

Appendix A

Overview of Model Development Approach

The modeling goal of the project is to predict the average delivery time (delivery_time_days) of each order on the Olist platform in order to assist enterprises in identifying possible delayed orders prior to their shipment so as to achieve more efficient logistics scheduling and customer satisfaction management. In model selection, we have clearly defined the model type as a regression prediction model as a result of the EDA that shows the distribution of features and nonlinear relationships. We hence adopted five typical regression models for comparative analysis covering a ground of different levels of complexity from linear models to ensemble models:

- **Multiple Linear Regression:** As the simplest and most interpretable model, this serves as the baseline model to set reference standards.
- **Lasso regression** (regularization model): While it maintains interpretability, it effectively reduces the occurence of multicollinearity by virtue of the introduction of a penalty term.
- **XGBoost Regression** (Extreme Gradient Boosting Regressor): An efficient tree model based on the concept of gradient boosting that captures nonlinear, complex relationships and high dimensional relations of the features.
- **Random Forest Regressor**: Integrating several trees reduces variance output and increases generalization ability.
- **LightGBM Regression** (Gradient Boosting Framework): As a representative of the family of gradient boosting models, it is characterized by effective iteration, inherently including feature importance mechanics, whereby an effective balance is struck between optimum performance and efficiency is obtained. In the process of evaluating the models generated we refer mainly to three core indices in order to ensure that the results obtained are comprehensive and reliable: whether the performance attained.

Quantum of business independencs:

- **MAE** (mean absolute error): an explainable indicator to express the bias of prediction.
- **RMSE** (root mean square error): a measure of the behaviour of the model indicative of its sensitivity to extremely delayed orders.
- **R^2** (coefficient of determination): an expression of the model's power to explain population variance.

Otherwise, in addition to technical performance, the degree of business interpretability and readiness for deployment of the model are also determinants of selection. Linear models are restricted in performance but provide visual representation of the contrapuntal influencing factors. Gradient boosting models like LightGBM and XGBoost are now at an ideal balance of prediction accuracy with inherent automation potential such as to be able to be integrated in the real-time decision-making processes of enterprises.

A.1 Baseline Development and Iteration

Prior to applying advanced models, the team implemented interpretable baselines to understand primary relationships between features related to logistics and time to deliver. Linear Regression was implemented in 2 iterations to increase model interpretability and predictive accuracy sequentially.

A.1.1 Simple Linear Regression (Pre–Feature Engineering)

The first model employed a simple Linear Regression framework with seven raw predictors based on the core logistical and transactional characteristics, such as freight value, product weight, volume and customer to seller distance. This model was designed to give a transparent benchmark of how good the basic data was in explaining the variability in delivery time without any transformations or features being generated.

Model	MAE	RMSE	R^2
Simple Linear Regression (No Feature Engineering)	5.4870	8.1205	0.1822

Table A1.1. Performance of Simple Linear Regression (5-Fold Cross Validation)

Though interpretable, this model had a poor fit, only accounting for 18% of the variance in delivery time. The weak fit indicated that there may be non-linear relationships as well as skewed distributions among characteristics such as cost of freight and distance to delivery.

A.1.2 Linear Regression with Feature Engineering

To overcome these limitations, a second baseline model was developed with targeted feature engineering. Logarithmic transformations were applied to reduce skewness in heavy-tailed variables (freight, distance, product dimensions), and region-based aggregations were introduced to capture spatial heterogeneity between customer and seller locations.

Model	MAE	RMSE	R^2
Linear Regression (With Feature Engineering)	5.1386	7.7239	0.2601

Table A1.2. Performance of Linear Regression with Feature Engineering (5-Fold CV)

This iteration achieved a noticeable improvement: MAE decreased by 6%, RMSE by 5%, and R^2 rose to 0.26. These results indicate that the transformed and engineered features effectively captured the underlying structure of the data.

A.1.3 Interpretation and Baseline Selection

A comparison of both versions of the model clearly indicates that the feature-engineered version of the model was able to outperform the raw baseline by a good margin, demonstrating improved explanatory power, robustness and interpretability, making it a suitable foundation for further modelling. This improved baseline was therefore adopted as the starting point for more complex models such as Lasso, LightGBM, XGBoost, and Random Forest, thus providing the methodological basis for the ensemble framework that is later developed in the project.

A.2 Lasso Regression

The second model is Lasso regression. When using linear regression, more features may affect the prediction. By using Lasso regression (L1 regularization), it adds a penalty term to avoid overfitting and select features.

For iteration, the first lasso regression model does not include Cross-validation and selecting hypermeter. The overall performance is shown below:

Model	RMSE	MAE	R^2
Lasso without hyperparameter and CV	7.8118	5.2277	0.2432
Lasso with hyperparameter and 5-Fold CV	7.7245	5.1389	0.2600

Lasso regression model is simpler and more efficient, it can be used to reduce computation cost. However, the overall performance is not better than Linear regression model based on metrics in Appendix C after hyperparameter turning. For hyperparameter selection, we use LassoCV to find the best penalty term.

A.3 LightGBM Models

The third modelling stage employed Light Gradient Boosting Machine (LightGBM), a decision-tree-based ensemble method designed to efficiently handle large-scale data and capture non-linear feature interactions. Unlike linear models that assume additive relationships between predictors and delivery time, LightGBM constructs a sequence of shallow trees that iteratively reduce residual errors, allowing it to represent complex dependencies among freight cost, geographical distance, and product characteristics.

A.3.1 Baseline LightGBM (No Tuning)

The first LightGBM experiment used default parameters to serve as a non-linear baseline, providing a benchmark against the previously tested Linear Regression models. A 5-Fold Cross-Validation framework was applied to ensure consistent evaluation and prevent overfitting.

Model	MAE	RMSE	R^2
LightGBM (Baseline, No Tuning)	4.9171	7.4792	0.3062

Table A3.1. Baseline LightGBM performance (5-Fold CV)

The untuned LightGBM already outperformed all linear models, reducing MAE from 5.14 days to 4.92 days and increasing R^2 from 0.26 to 0.31. This confirms that delivery dynamics are inherently non-linear — particularly the long-tail effects of customer–seller distance and freight cost.

Feature-importance analysis revealed that the log-transformed freight value, city-level mean delivery times, and log-scaled product dimensions were the strongest predictors. This result aligns closely with the EDA findings, reinforcing that delivery speed is primarily driven by logistics and geographic attributes rather than product or payment characteristics.

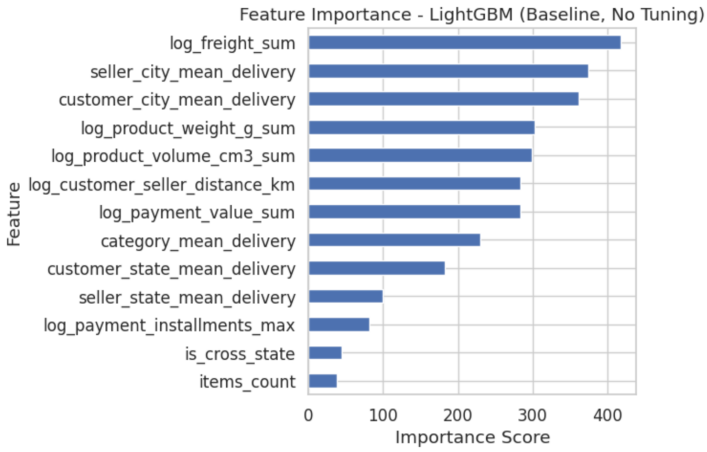


Figure A3.1. Feature Importance – LightGBM (Baseline, No Tuning)

A.3.2 LightGBM with Hyperparameter Optimisation (Optuna)

To further enhance model accuracy and generalisability, a second iteration was conducted using Optuna to perform Bayesian hyperparameter optimisation. The objective function minimised cross-validated RMSE across five folds, tuning the following parameters: learning_rate, num_leaves, max_depth, subsample, colsample_bytree, min_child_samples, reg_alpha, and reg_lambda.

This tuning process was designed to control overfitting while improving the model's ability to adapt to heterogeneous delivery scenarios across Brazilian regions. The final tuned model achieved the following performance:

Model	MAE	RMSE	R^2
LightGBM (Baseline, No Tuning)	4.9171	7.4792	0.3062
LightGBM (Optuna Tuned, 5-Fold CV)	4.8660	7.4277	0.3158

Table A3.2. Performance comparison before and after hyperparameter tuning

The tuning process produced a modest yet consistent improvement: MAE decreased by approximately 1.0%, RMSE by 0.7%, and R^2 increased to 0.316. Though incremental, this uplift validates that even light-weight optimisation can refine model stability and predictive precision without increasing computational cost substantially.

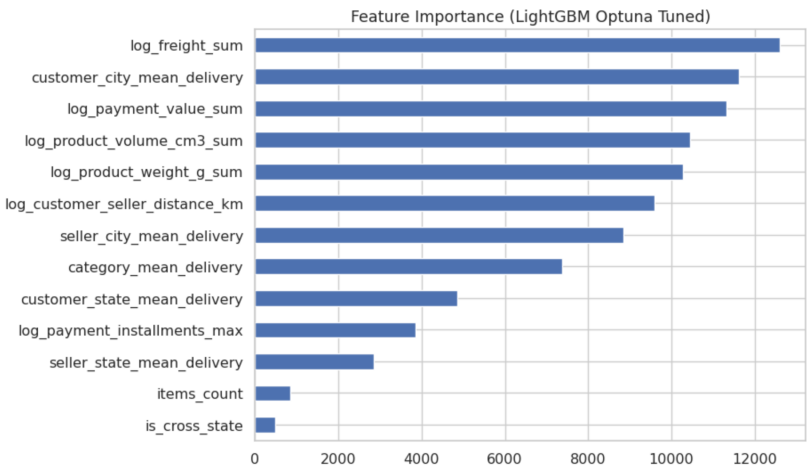


Figure A3.2. Feature Importance – LightGBM (Optuna Tuned)

The optimised model retained similar top predictors — log_freight_sum, customer_city_mean_delivery, and log_payment_value_sum — demonstrating that the model's interpretability remains stable across iterations. Minor changes in feature ranking suggest that the tuning improved balance between city-level and product-level variables, enhancing its generalisation across diverse delivery conditions.

A.3.3 Interpretation and Role in the Modelling Pipeline

The LightGBM model proved to be a powerful intermediary step between interpretable linear methods and more complex ensemble approaches. Its ability to model non-linear, interaction-driven effects—particularly those related to distance and freight—makes it ideal for integration into the final stacking ensemble.

Moreover, LightGBM's feature-importance outputs provide business-relevant insights: for example, freight value and inter-city distance are repeatedly confirmed as dominant drivers of delay, suggesting that Olist should prioritise these metrics when optimising routing and warehouse allocation.

Overall, this stage demonstrates a clear and disciplined experimentation path: Start with interpretable linear baselines to build intuition. Introduce tree-based methods to capture non-linear delivery dynamics. Apply targeted hyperparameter tuning to maximise predictive efficiency. Use the optimised LightGBM as a high-performance component in the final stacked model.

This systematic refinement not only improves predictive accuracy but also strengthens the practical interpretability and deployability of Olist's predictive delivery-time framework.

A.4 XGBoost Models

The next stage of model development applied Extreme Gradient Boosting (XGBoost), a gradient-boosted decision-tree algorithm known for its robustness, scalability, and strong performance on structured tabular data.

While similar in concept to LightGBM, XGBoost uses a level-wise tree growth strategy and integrates regularisation terms to prevent overfitting, making it a valuable benchmark for non-linear prediction in logistics analytics.

A.4.1 Baseline XGBoost (No Tuning)

The baseline experiment implemented XGBoost with default hyperparameters to assess its performance under a simple configuration. Using 5-Fold Cross-Validation, the model achieved the following results:

Model	MAE	RMSE	R^2
XGBoost (Baseline, No Tuning)	4.9485	7.5422	0.2945

Table A4.1. XGBoost baseline model performance (5-Fold CV)

The baseline XGBoost already outperformed the linear models and achieved similar accuracy to LightGBM, confirming that gradient-boosted trees are well-suited to capturing the complex non-linear effects in delivery-time data.

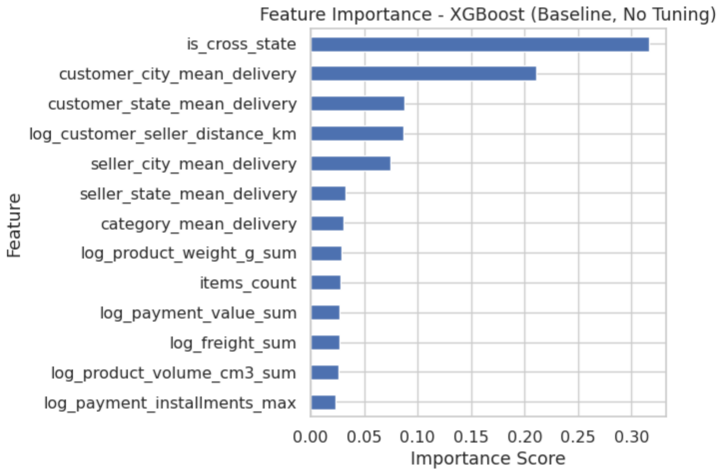


Figure A4.1. Feature Importance – XGBoost (Baseline, No Tuning)

Feature-importance analysis highlights is_cross_state, customer_city_mean_delivery, and customer_state_mean_delivery as the top predictors, underscoring the central role of regional and cross-state logistics factors in determining delivery duration. This finding supports Olist's business insight that geographical segmentation heavily influences shipping efficiency.

A.4.2 XGBoost with Hyperparameter Optimisation (Optuna)

To further refine model performance and generalisability, Optuna was employed for Bayesian hyperparameter optimisation.

The objective function minimised RMSE across 5-fold cross-validation, tuning parameters related to model depth, learning rate, subsampling, and regularisation:

Key Parameters Tuned	
learning_rate	0.0249
max_depth	7
subsample	0.7728
colsample_bytree	0.7165
min_child_weight	6.5067
gamma	0.1395
reg_alpha	0.5843
reg_lambda	0.7327

Table A4.1Key Parameters Tuned

The optimised configuration achieved measurable improvements:

Model	MAE	RMSE	R^2
XGBoost (Baseline, No Tuning)	4.9485	7.5422	0.2945
XGBoost (Optuna Tuned, 5-Fold CV)	4.8611	7.4258	0.3161

Table A4.2. Performance comparison of XGBoost before and after tuning

The Optuna-tuned model reduced both MAE and RMSE while improving the R^2 from 0.29 to 0.32, indicating better model generalisation and stability.

The most influential features after tuning (Figure A4.2) include log_freight_sum, customer_city_mean_delivery, and log_payment_value_sum, confirming that cost-related and city-level delivery characteristics are consistently strong predictors across models.

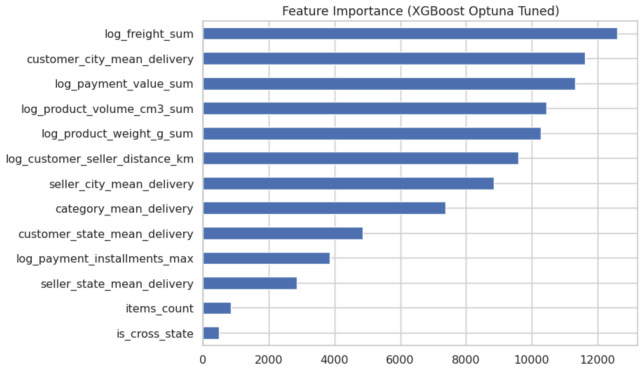


Figure A4.2. Feature Importance – XGBoost (Optuna Tuned)

The optimisation also led to a more balanced feature weighting, where both product-specific (e.g., log_product_volume_cm3_sum) and spatial variables (e.g., log_customer_seller_distance_km) contribute meaningfully to model output. This balance reflects XGBoost's ability to simultaneously model local delivery nuances and broad logistical patterns.

A.4.3 Interpretation and Integration

The XGBoost results demonstrate that even small-scale tuning significantly enhances model accuracy without compromising interpretability. Compared with LightGBM, XGBoost exhibits slightly higher regularisation strength and greater stability across folds, making it an excellent complement within an ensemble framework.

These insights guided the next modelling phase — **the Stacking Ensemble**, where XGBoost (as a strong non-linear learner) is combined with Linear Regression and LightGBM to produce an optimised, hybrid model that maximises predictive performance and robustness.

A.5 Random Forest Models

The Random Forest Regression model was implemented as a non-linear ensemble method to capture complex interactions between spatial, logistical, and transactional variables affecting delivery time. By aggregating multiple independent decision trees trained on random subsets of data and features, Random Forest reduces overfitting and improves predictive stability. Two model iterations were developed to evaluate performance progression — a baseline version using default parameters and an optimised version using Optuna-based hyperparameter tuning.

A.5.1 Baseline Random Forest (No Tuning)

The baseline Random Forest served as the initial non-linear benchmark following the linear and gradient boosting experiments. It was constructed using default parameters with a small number of trees (n_estimators=10) and no depth constraints. The model was evaluated through 5-Fold Cross Validation to ensure robustness:

Model	MAE	RMSE	R^2
Random Forest (Baseline, No Tuning)	5.3384	7.9863	0.2090

Table A5.1. Baseline Random Forest performance (5-Fold CV)

While the baseline Random Forest demonstrated higher accuracy than the simplest Linear Regression models, its relatively low R^2 (0.21) suggested underfitting due to limited depth and ensemble size. The small number of estimators restricted the model's ability to capture complex non-linear effects, particularly in cases of long-distance deliveries and heavy parcels.

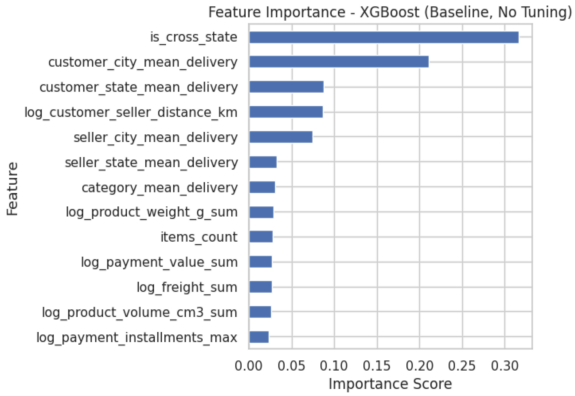


Figure A5.1. Feature Importance – Random Forest (Baseline, No Tuning)

Feature importance analysis revealed that customer_seller_distance_km, customer_city_mean_delivery, and freight_sum were the most influential features. This result aligned with prior EDA findings, highlighting the strong spatial dependency in delivery-time variation. However, several low-ranked features showed minimal contribution, indicating potential redundancy and limited sampling diversity under the default configuration.

A.5.2 Random Forest with Hyperparameter Optimisation (Optuna)

To enhance predictive capacity and reduce variance, a second iteration of the model employed Optuna for hyperparameter tuning. The optimisation targeted the bias–variance trade-off by adjusting parameters including:

- n_estimators (number of trees: 100–300)
- max_depth (tree depth: 5–15)
- min_samples_split and min_samples_leaf (node-splitting thresholds)
- max_features (sqrt vs log2)
- bootstrap (sampling strategy)

The objective function minimised cross-validated RMSE across five folds, exploring five trials to identify a stable configuration. The final model used n_estimators=175, max_depth=15, min_samples_split=8, min_samples_leaf=3, and max_features='sqrt'.

Model	MAE	RMSE	R^2

Random Forest (Baseline, No Tuning)	5.3384	7.9863	0.2090
Random Forest (Optuna Tuned, 5-Fold CV)	4.9012	7.4654	0.3088

Table A5.2. Performance comparison before and after hyperparameter tuning

The tuned Random Forest showed significant improvement in all evaluation metrics: MAE dropped by approximately 8%, RMSE decreased by 6.5%, and R^2 rose to 0.31, confirming stronger model fit and generalisation.

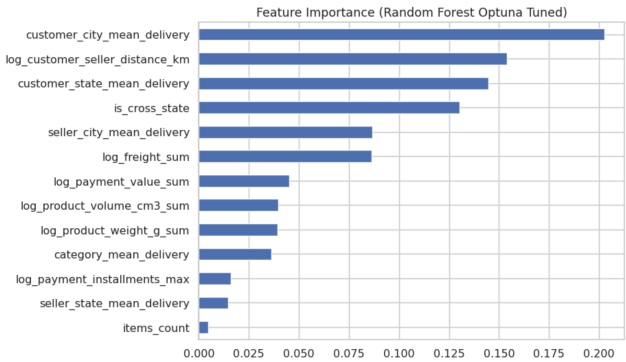


Figure A5.2. Feature Importance – Random Forest (Optuna Tuned)

After tuning, the feature-importance distribution became more balanced. customer_city_mean_delivery and customer_state_mean_delivery remained dominant, while engineered variables like log_freight_sum and is_cross_state gained greater influence. This reflects the model's enhanced ability to integrate local and cross-regional delivery dynamics.

A.5.3 Model Iteration and Interpretation

The iterative process from the baseline to the tuned Random Forest demonstrates a clear improvement in predictive robustness and representational depth.

The baseline model established a simple, stable foundation, while the Optuna-tuned version achieved stronger generalisation by balancing depth and sampling diversity.

Compared with boosting models (LightGBM and XGBoost), Random Forest achieved slightly lower peak accuracy but delivered more stable results across folds and remained less sensitive to parameter noise.

Together, these results validate Random Forest as a crucial non-linear learner in the modelling pipeline — one that bridges interpretability and performance and complements the gradient boosting algorithms within the final ensemble framework.

Conclusion & Final Model Recommendations

From the above results, the XGBoost and LightGBM models tuned by Optuna have achieved the best performance in both RMSE and MAE. After comparison, we chose both as our best single model. From the perspective of computational cost, the computational cost of the random forest algorithm is very high. Linear regression and Lasso regression performed relatively poorly.

We finally adopted the Stacking Ensemble model, taking the optimized XGBoost & LightGBM as the base learner and linear regression as the meta-learner. The performance of this model is as follows:

Model	MAE	RMSE	R^2
Stacking (XGBoost & LightGBM + Linear Meta)	4.0439	6.7856	0.4289

It can be seen from the results that although the improvement of the Stacking model is limited, it has certain advantages in stability and generalization ability. To further enhance the model consistency and non-linear fitting capability, we replaced the linear meta-learner with a LightGBM meta-model in the stacking framework. The LightGBM meta-learner would better capture complex non-linear relationships among base model outputs, resulting in slightly improved MAE and RMSE performance while maintaining stable generalization. The result of model performance is shown below:

Model	MAE	RMSE	R^2
Stacking (XGBoost & LightGBM + LightGBM Meta)	3.6457	6.3413	0.5013

In conclusion, this model not only performs stably in terms of prediction accuracy but also retains the interpretability of some linear layers, facilitating enterprises to identify key influencing factors (such as transportation distance, products weight, regional differences, etc.) and be used for dynamic scheduling and customer expectation management in actual operations.